

Module 4

What is multiplexing? What are the different types of multiplexing? Explain TDM with a neat block diagram.

Define pulse amplitude modulation. Obtain the expression for the Fourier transform of PAM signal.

Define pulse amplitude modulation. Obtain the expression for Fourier transform of PAM with spectrum diagram

What is quantization process? Explain the different types of Quantization with their input output characteristics.

What is quantization noise ? Derive the output signal to noise ratio of quantizer

With neat diagram explain the generation and detection of PPM waves

Explain the effect of noise in Pulse Position modulation

With relevant diagram explain the generation and reconstruction of pulse code modulation (PCM) signal

State and prove sampling theorem for bandlimited signals with waveforms and equations

What are the advantages of digital signals over analog signals

Distinguish between PAM,PCM PPM

Explain undersampling,oversampling and Nyquist rate

What is quantization noise. Show that signal to noise ratio of uniform quantizer is equal to $1.8+6N$

What is aperture effect in PAM systems? How can it be minimized?

Develop a program OF sampling theorem to display the signals and its spectrum

L3

What are the desirable properties of digital waveforms? To transmit a bit sequence 10011011, draw the resulting waveforms using:- Unipolar NRZ; polar NRZ; Unipolar RZ ; Bipolar RZ ; Manchester(split phase)

PCM system uses uniform quantizer followed by a 7 bit binary encoder. The bit rate of the system is 50×10^6 bps. What is minimum message bandwidth?

A TV signal with a bandwidth of 4.2 MHz is transmitted using binary PCM. The number of representation level is 512. Calculate:

- (i) Codeword length
- (ii) Final bit rate
- (iii) Transmission bandwidth

An audio signal digitalized using PCM. Assume the audio signal bandwidth to be 20kHz.

- i) What is the Nyquist rate and Nyquist period of the audio signal?
- ii) If the samples are quantized to $L_s = 4096$ levels and then binary coded, determine the number of bits required to encode a sample.

Draw the line codes for given binary representation 01101001

- i) Unipolar NRZ signaling
- ii) Polar NRZ signaling
- iii) Unipolar RZ signaling
- iv) Bipolar RZ signaling
- v) Manchester code

Determine the Nyquist rate and the Nyquist interval for :

- (i) $g(t) = \sin c(200t)$
- (ii) $m(t) = \frac{1}{\pi t} [\sin(500\pi t)]$.